

Standalone Probit

probit family, scored against the market on 2015-2026, 5 seed(s), from unrecorded commit

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Model identity and provenance

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Standalone Probit probit	MEASURED AGAINST the market the same yardstick in all four reports	SEEDS 5 7, 17, 42, 123, 256	TEST YEARS 12 2015-2026
BOOSTING ROUNDS 400 3-fold CV	TRAINED ON every earlier season, expanding the mean learner is refitted per test year on every season before it -- 12 fits in all	SOURCE COMMIT n/a clean	

These are walk-forward figures, not one model's: the softmax mean learner is refitted for each test year on every season in the data before it, so the result concatenates 12 separately fitted models and no test year appears in its own training data. The window starts at the first season the archive holds rather than at the run config's start year, and this report does not read the feature table, so it does not name that year.

Config digest unrecorded, model view bagged_probit, artifacts read from results/probit_models.

ENVIRONMENT

Component	Version
numpy	2.5.2
pandas	3.0.5
python	3.12.11
xgboost	3.4.1

Deltas in this report use `model_minus_market_log_loss`: `race_multinomial_log_loss`, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Executive summary

MEAN DELTA VS MARKET +0.002018 negative is better	RACES 8,675	YEARS BETTER 4/11	ISOLATION DELTA -0.012557 against the comparator it was designed against
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Standalone Probit is worse than the market by 0.002018 race_multinomial_log_loss on 8,675 races. The direction holds in 4 of 11 evaluated years. It is behind the market: on the common yardstick this model loses to the prices it was measured against. Against the standalone Softmax on the same utilities -- the comparison this architecture was designed to make, which is the only one that isolates the link change -- the delta is -0.012557. That is the opposite sign to the market delta above: the candidate beats what it was built to beat and still loses to the market, and both are true. Every seed agrees on the sign of that isolation delta.

Comparator and evaluation population

The market's own implied probabilities on the same races. Every published report is scored against the market, so the four headline numbers mean the same thing and can be read against one another. For the production model the market is also the baseline by construction -- the booster learns a residual on log(market_prob) -- and for the probit candidates it is the common yardstick rather than the quantity they were designed against; the comparison each was designed against is in the technical appendix.

TEST RACES
8,675

YEARS
12
2015-2026

SEEDS
5

HONEST-SCALE YEARS
11
years whose utility scale
was fitted rather than
fallen back

Some years fell back rather than fitting their own scale

2015 did not produce an honest utility scale (insufficient_history). Those years are excluded from the headline delta and from the bankroll paths, and shown separately in the year-by-year table: a fallback scale is not a measurement of the candidate.

Overall probability performance

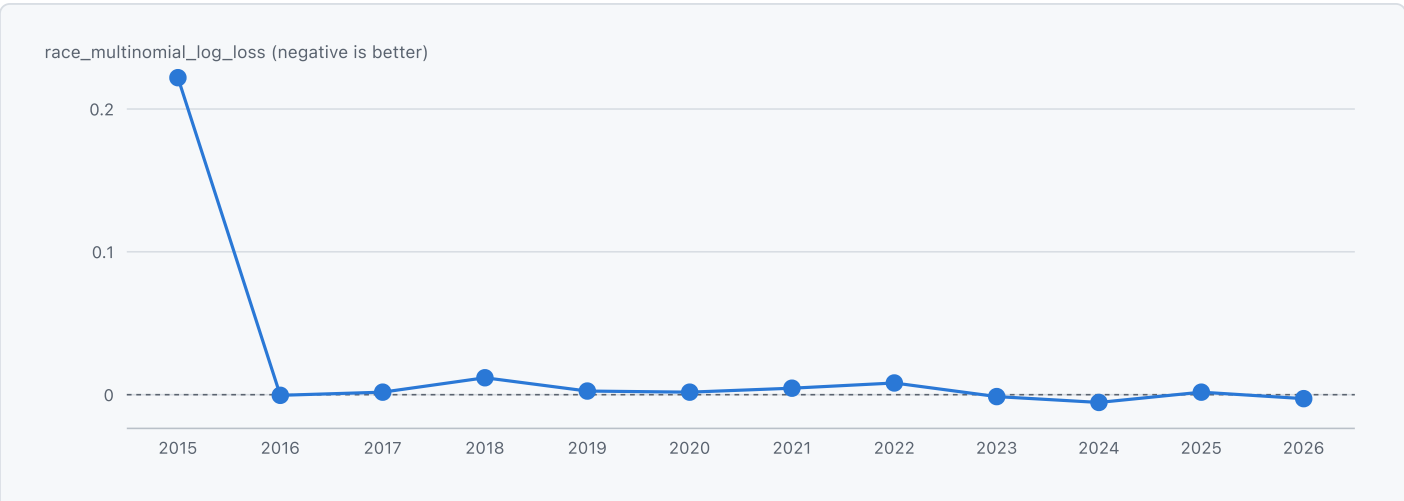
races	model log loss	market log loss	market delta	better year share
8675	2.030823	2.028805	0.002018	0.363636

One row, pooled over every evaluated race, and measured against the market: market_delta is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	market delta	honest
2015	742	2.220620	1.998720	0.221900	no
2016	744	2.098827	2.099277	-0.000450	yes
2017	772	2.023286	2.021559	0.001727	yes
2018	756	2.073354	2.061504	0.011849	yes
2019	759	1.935353	1.932827	0.002526	yes
2020	799	2.012171	2.010400	0.001771	yes
2021	783	2.011146	2.006659	0.004487	yes
2022	783	2.059349	2.051195	0.008154	yes

test year	races	model log loss	market log loss	market delta	honest
2023	796	2.022700	2.024066	-0.001366	yes
2024	792	2.020289	2.025744	-0.005455	yes
2025	748	2.059159	2.057397	0.001763	yes
2026	201	2.023421	2.026224	-0.002803	yes



Mean delta by year Mean delta against the market for each evaluated year. Zero is the market.

▼ Table view

Category	race_multinomial_log_loss (negative is better)
2015	0.2219
2016	-0.00045
2017	0.001727
2018	0.011849
2019	0.002526
2020	0.001771
2021	0.004487
2022	0.008154
2023	-0.001366
2024	-0.005455
2025	0.001763
2026	-0.002803

Ahead of the market in 4 of 12 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

This architecture reports calibration through a paired comparison rather than binned reliability: candidate and comparator are each given a temperature on the same races, so the calibrated delta is the like-for-like figure. Fallback-scale years are excluded here, as they are from the headline.

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2016	2.096131	2.103251	-0.007120
2017	2.024009	2.031793	-0.007785
2018	2.073354	2.079390	-0.006036
2019	1.938096	1.945433	-0.007337
2020	2.012574	2.024462	-0.011888

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2021	2.011085	2.017480	-0.006395
2022	2.059959	2.065483	-0.005524
2023	2.023223	2.025731	-0.002508
2024	2.021114	2.026537	-0.005423
2025	2.060147	2.066830	-0.006683
2026	2.024664	2.035721	-0.011056

If the calibrated delta is materially smaller than the uncalibrated one, part of the candidate's advantage was a temperature the comparator could also have had.

Top-choice movement

Not applicable to this model. Top-choice movement compares a model's first pick against the market's. This candidate is scored against its own comparator on fixed mean utilities, not against the market, so a market top-choice comparison would answer a different question than the rest of this report.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable. This architecture is a choice *link* applied to one set of mean utilities, not a bag of independently fitted members, so seed variation here is integration noise rather than model disagreement -- and that is reported in the appendix instead.

Feature and correction effects

Not applicable to this model. There are no per-feature effects belonging to this model to attribute. A choice link changes how utilities become probabilities; it has no features of its own. The mean utilities come from the comparator's fit, so attributing their feature effects here would credit this candidate with another model's work -- and for the market-anchored architectures those are the incumbent Offset's margins exactly, whose per-feature effects, response directions and group ablation are in the Bagged Offset report. What is specific to this candidate is the link and its utility scale, and both are in the technical appendix.

Representative races

Not applicable to this model. Representative races are chosen by how much a model's correction moved a runner's probability. This candidate applies a link to fixed mean utilities, so the movement to explain belongs to the comparator's fit and is shown in that model's report. A race chosen here would be chosen by integration noise.

Betting and bankroll analysis

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

SETTLEMENT BY STAKING RULE

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
flat_stake@edge>=0.02	6510	65,100,000	-25,665,000	-0.394240	-36.841361	2.531206	-15,665,000	3833
kelly_no_rebate@0.01	7363	15,537,960	-2145786.500000	-0.138100	-0.241535	0.221390	7854213.500000	4256
kelly_no_rebate@0.02	7399	28,029,580	-3,860,391	-0.137726	-0.487824	0.396485	6,139,609	4275
kelly_no_rebate@0.05	7407	52,710,070	-7,124,518	-0.135164	-1.246365	0.723988	2,875,482	4282
kelly_no_rebate@0.1	7377	71,427,330	-9,245,495	-0.129439	-2.584278	0.930033	754,505	4262
kelly_settled_rebate@0.01	7369	16,228,700	-2,254,076	-0.138894	-0.162460	0.161965	8,500,497	4254
kelly_settled_rebate@0.02	7405	30,525,200	-4,236,205	-0.138777	-0.323747	0.296708	7,234,332	4275
kelly_settled_rebate@0.05	7418	63,771,520	-8807677.500000	-0.138113	-0.823451	0.590323	4389142.500000	4289
kelly_settled_rebate@0.1	7414	96,951,250	-13,224,599	-0.136405	-1.728009	0.844323	1,776,377	4286
rebate_aware_kelly@0.01	7715	18,207,680	-2,607,460	-0.143207	-0.191536	0.188440	8256902.450000	4251
rebate_aware_kelly@0.02	7815	35,887,720	-5,181,461	-0.144380	-0.403814	0.352974	6677686.250000	4280
rebate_aware_kelly@0.05	8054	76,249,900	-10236084.500000	-0.134244	-0.935557	0.636599	3923673.600000	4402
rebate_aware_kelly@0.1	7937	105,491,460	-14313770.500000	-0.135687	-2.055409	0.889349	1280404.400000	4352

Model view bagged_probit. Seed -1. Every rule is settled on the same selected bets at the same odds, so the differences between these rows are stake sizing alone.

What this settlement can decide

Secondary diagnostic. This suite selects on paired race log loss; betting returns cannot select a model and are not evidence for or against a probit candidate.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.034377	0.034377	-0.1738	951
0.050000 to 0.100000	0.072830	0.072830	0.0143	1002
0.100000 to 0.200000	0.144020	0.144020	-0.3538	1230
0.200000 to unbounded	0.578110	0.578110	-0.5952	3327

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.010508	0.010508	-0.2136	-0.2361	-0.2906	-0.4206	853	889	897	867
0.020000 to 0.050000	0.034377	0.034377	-0.0045	-0.0227	-0.0838	-0.2033	951	951	951	951
0.050000 to 0.100000	0.072830	0.072830	-0.2092	-0.2113	-0.2121	-0.1886	1002	1002	1002	1002
0.100000 to 0.200000	0.144020	0.144020	-0.0966	-0.0846	-0.0509	-0.0084	1230	1230	1230	1230
0.200000 to unbounded	0.578110	0.578110	-0.1469	-0.1472	-0.1465	-0.1431	3327	3327	3327	3327

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.010483	0.010483	-0.2098	-0.2203	-0.2503	-0.3224	859	895	908	904
0.020000 to 0.050000	0.034377	0.034377	0.0008	-0.0120	-0.0512	-0.1307	951	951	951	951
0.050000 to 0.100000	0.072830	0.072830	-0.2098	-0.2105	-0.2141	-0.2102	1002	1002	1002	1002
0.100000 to 0.200000	0.144020	0.144020	-0.1008	-0.0930	-0.0693	-0.0338	1230	1230	1230	1230
0.200000 to unbounded	0.578110	0.578110	-0.1475	-0.1479	-0.1489	-0.1495	3327	3327	3327	3327

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.031497	-0.031497	-0.3107	-0.3763	-0.1766	-0.2306	351	412	643	531
0 to 0.020000	0.010503	0.010503	-0.2290	-0.1475	-0.1207	-0.1626	854	893	901	896
0.020000 to 0.050000	0.034377	0.034377	0.0163	-0.0149	-0.0791	-0.1452	951	951	951	951
0.050000 to 0.100000	0.072830	0.072830	-0.2118	-0.1868	-0.1819	-0.1980	1002	1002	1002	1002
0.100000 to 0.200000	0.144020	0.144020	-0.0853	-0.0891	-0.0686	-0.0368	1230	1230	1230	1230

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0.200000 to unbounded	0.578110	0.578110	-0.1488	-0.1504	-0.1483	-0.1447	3327	3327	3327	3327

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL HK\$10,000,000	PRESETS SETTLED 1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly	PATH MODES annual reset, continuous	SEEDS 5 7, 17, 42, 123, 256
TRAINED ON every earlier season, expanding the mean learner is refitted per test year on every season before it -- 12 fits in all	BOOSTING ROUNDS 400	YEARS STAKED 2015-2026	

These are walk-forward figures, not one model's: the softmax mean learner is refitted for each test year on every season in the data before it, so the result concatenates 12 separately fitted models and no test year appears in its own training data. The window starts at the first season the archive holds rather than at the run config's start year, and this report does not read the feature table, so it does not name that year. Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Where a year is excluded, a continuous path opens at the balance the bank held when the kept years begin and an annual-reset path opens at the declared bankroll, so the two show different opening figures for the same rule. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. 2015 is excluded from every path below. These test years have no earlier year to fit the utility scale on and fall back to the predeclared theoretical value. The utility is divided by the scale, so the fallback amplifies corrections where a fitted scale halves them: their stakes and claimed edges are several times any fitted year's and are not comparable with them. Exclude these years from any return figure -- filter on `is_honest_scale` in the per-run CSV.

1% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

-0.116%

Kelly, no rebate, annual reset

BETS

6,121

settled bets

REBATE RECEIVED

HK\$62,394

the most any rule received

■ Kelly, no rebate ■ Kelly, settled rebate ■ Rebate-aware Kelly



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	3,525	-0.116%	6,121 bets
Kelly, no rebate, continuous	3,525	-6.3%	6,121 bets
Kelly, settled rebate, annual reset	3,528	-0.125%	6,127 bets
Kelly, settled rebate, continuous	3,528	-6.3%	6,127 bets
Rebate-aware Kelly, annual reset	3,525	-0.122%	6,283 bets
Rebate-aware Kelly, continuous	3,525	-7.2%	6,283 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	751	751	773
2017	881	883	911
2018	789	789	811
2019	1,038	1,038	1,068
2020	964	965	980
2021	570	571	583
2022	279	279	285
2023	230	230	236
2024	298	299	308
2025	261	261	267

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2026	60	61	61

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 751 bets in 2016 against 261 in 2025. Kelly, settled rebate: 751 bets in 2016 against 261 in 2025. Rebate-aware Kelly: 773 bets in 2016 against 267 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$62,394; Rebate-aware Kelly, continuous HK\$62,394; Kelly, settled rebate, annual reset HK\$4,864; Kelly, settled rebate, continuous HK\$4,864. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

2% KELLY

OPENING BANKROLL
HK\$10,000,000

BEST FINAL RETURN
-0.182%
Kelly, no rebate, annual reset

BETS
6,157
settled bets

REBATE RECEIVED
HK\$340,791
the most any rule received

Kelly, no rebate
Kelly, settled rebate
Rebate-aware Kelly



Cumulative bankroll return at 2% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	3,544	-0.182%	6,157 bets
Kelly, no rebate, continuous	3,544	-12.4%	6,157 bets
Kelly, settled rebate, annual reset	3,549	-0.214%	6,163 bets
Kelly, settled rebate, continuous	3,549	-12.1%	6,163 bets
Rebate-aware Kelly, annual reset	3,553	-0.464%	6,383 bets
Rebate-aware Kelly, continuous	3,553	-15.0%	6,383 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	754	754	790
2017	886	886	921
2018	794	795	832
2019	1,040	1,041	1,071
2020	968	968	993
2021	574	575	591
2022	285	285	295
2023	231	231	237
2024	302	305	316
2025	262	262	275
2026	61	61	62

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 754 bets in 2016 against 262 in 2025. Kelly, settled rebate: 754 bets in 2016 against 262 in 2025. Rebate-aware Kelly: 790 bets in 2016 against 275 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$340,791; Rebate-aware Kelly, continuous HK\$340,791; Kelly, settled rebate, annual reset HK\$28,170; Kelly, settled rebate, continuous HK\$28,170. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

5% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

-0.213%

Kelly, no rebate, annual reset

BETS

6,165

settled bets

REBATE RECEIVED

HK\$965,345

the most any rule received

Kelly, no rebate Kelly, settled rebate Rebate-aware Kelly



Cumulative bankroll return at 5% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	3,551	-0.213%	6,165 bets
Kelly, no rebate, continuous	3,551	-28.4%	6,165 bets
Kelly, settled rebate, annual reset	3,562	-0.325%	6,176 bets
Kelly, settled rebate, continuous	3,562	-27.0%	6,176 bets
Rebate-aware Kelly, annual reset	3,675	-0.524%	6,622 bets
Rebate-aware Kelly, continuous	3,675	-26.6%	6,622 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	754	754	849
2017	886	888	972
2018	796	797	862
2019	1,041	1,043	1,087
2020	968	968	1,006
2021	576	579	612
2022	285	287	311
2023	231	231	252
2024	305	305	324
2025	262	262	280
2026	61	62	67

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 754 bets in 2016 against 262 in 2025. Kelly, settled rebate: 754 bets in 2016 against 262 in 2025. Rebate-aware Kelly: 849 bets in 2016 against 280 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$965,345; Rebate-aware Kelly, continuous HK\$965,345; Kelly, settled rebate, annual reset HK\$96,858; Kelly, settled rebate, continuous HK\$96,858. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds.

Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

10% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

-0.112%

Kelly, no rebate, annual reset

BETS

6,135

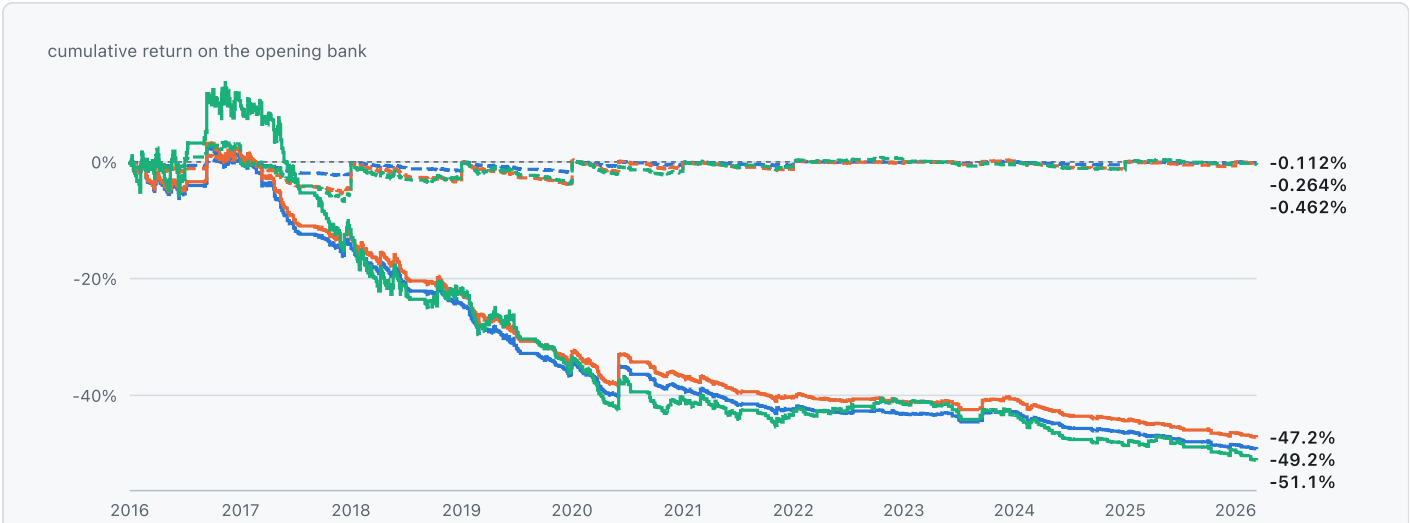
settled bets

REBATE RECEIVED

HK\$684,931

the most any rule received

■ Kelly, no rebate ■ Kelly, settled rebate ■ Rebate-aware Kelly



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	3,531	-0.112%	6,135 bets
Kelly, no rebate, continuous	3,531	-49.2%	6,135 bets
Kelly, settled rebate, annual reset	3,558	-0.264%	6,172 bets
Kelly, settled rebate, continuous	3,558	-47.2%	6,172 bets
Rebate-aware Kelly, annual reset	3,625	-0.462%	6,505 bets
Rebate-aware Kelly, continuous	3,625	-51.1%	6,505 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	754	754	849
2017	885	888	962
2018	793	797	846
2019	1,039	1,041	1,076
2020	965	968	994
2021	571	578	595
2022	279	287	295

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2023	230	231	238
2024	298	305	317
2025	261	262	271
2026	60	61	62

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 754 bets in 2016 against 261 in 2025. Kelly, settled rebate: 754 bets in 2016 against 262 in 2025. Rebate-aware Kelly: 849 bets in 2016 against 271 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$684,931; Rebate-aware Kelly, continuous HK\$684,931; Kelly, settled rebate, annual reset HK\$97,210; Kelly, settled rebate, continuous HK\$97,210. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. No staking rule at this fraction converts the model's edge into money on these odds. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,988,420	-0.116%	6,121	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$8,385,931	HK\$7,854,214	-6.3%	6,121	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,987,516	-0.125%	6,127	HK\$4,864
1% Kelly	Kelly, settled rebate, continuous	HK\$9,076,780	HK\$8,500,497	-6.3%	6,127	HK\$4,864
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,987,824	-0.122%	6,283	HK\$62,394
1% Kelly	Rebate-aware Kelly, continuous	HK\$8,897,664	HK\$8,256,902	-7.2%	6,283	HK\$62,394
2% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,981,834	-0.182%	6,157	HK\$0
2% Kelly	Kelly, no rebate, continuous	HK\$7,009,287	HK\$6,139,609	-12.4%	6,157	HK\$0
2% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,978,642	-0.214%	6,163	HK\$28,170
2% Kelly	Kelly, settled rebate, continuous	HK\$8,231,586	HK\$7,234,332	-12.1%	6,163	HK\$28,170
2% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,953,593	-0.464%	6,383	HK\$340,791
2% Kelly	Rebate-aware Kelly, continuous	HK\$7,857,632	HK\$6,677,686	-15.0%	6,383	HK\$340,791
5% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,978,690	-0.213%	6,165	HK\$0
5% Kelly	Kelly, no rebate, continuous	HK\$4,013,972	HK\$2,875,482	-28.4%	6,165	HK\$0
5% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,967,520	-0.325%	6,176	HK\$96,858
5% Kelly	Kelly, settled rebate, continuous	HK\$6,015,459	HK\$4,389,142	-27.0%	6,176	HK\$96,858
5% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,947,649	-0.524%	6,622	HK\$965,345
5% Kelly	Rebate-aware Kelly, continuous	HK\$5,347,723	HK\$3,923,674	-26.6%	6,622	HK\$965,345
10% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$9,988,758	-0.112%	6,135	HK\$0
10% Kelly	Kelly, no rebate, continuous	HK\$1,485,870	HK\$754,505	-49.2%	6,135	HK\$0
10% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$9,973,568	-0.264%	6,172	HK\$97,210
10% Kelly	Kelly, settled rebate, continuous	HK\$3,364,848	HK\$1,776,377	-47.2%	6,172	HK\$97,210
10% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,953,850	-0.462%	6,505	HK\$684,931
10% Kelly	Rebate-aware Kelly, continuous	HK\$2,620,569	HK\$1,280,404	-51.1%	6,505	HK\$684,931

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

MEAN DELTA
-0.012063

INTERVAL
-0.0140 to
-0.0101
80% level

EXCLUDES ZERO
FAVOURABLY
yes

RACE DAYS
891
39,665 races

REPLICATES
2,000
seed 20260824

A day-block bootstrap: race days are resampled within each test year, so the interval accounts for clustering within a meeting, and the interval lies entirely below zero. It conditions on the fitted model and does *not* include refit uncertainty.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix

THE UTILITY SCALE

FITTED MINIMUM
1.3893

FITTED MAXIMUM
1.4534

THEORY
0.7797
the value theory predicts

The fitted scale sits above the value theory predicts, in every year: 1.389 to 1.453 against 0.780 (the scale the fallback year used, which is the predeclared theoretical value). Theory gets this wrong and the fit is what the report uses -- a scale asserted from theory would divide the utilities by too little, and at that value the market-relative candidate loses instead of winning. A run that took the theoretical value on faith would have reported the wrong sign, which is why the first year -- which has no history and falls back to it -- is excluded from the headline and labelled.

CONFIGURATION

Item	Value
Architecture	probit
Comparator field	softmax
Integration	exact_gauss_hermite_quadrature
Shared factors	0
Opening bankroll	10,000,000
Bankroll mode	continuous
Rebate-aware sizing	no

THE COMPARISON THIS ARCHITECTURE WAS DESIGNED TO MAKE

ISOLATION COMPARATOR

the standalone
Softmax on the
same utilities

DELTA

-0.012557

negative is better

CALIBRATED

-0.007069

both sides given a
temperature

YEARS BETTER

11/11

WORST YEAR

-0.006424

Softmax, holding the mean utilities fixed. That is the only comparison that isolates the link: does replacing independent Gumbel noise with correlated normal noise help? Scoring the candidate against the Offset model instead would change the mean learner and the link at the same time. This report's headline is the market delta, because that is the one quantity all four published models can be compared on. This delta answers a narrower question and answers it better: it is the only comparison that holds everything except the link fixed. Read as a headline it would mislead -- a candidate can beat what it was designed against and still lose to the market -- and read as an isolation result it is exactly what the architecture was built to measure.

PROMOTION GATES

Gate	Verdict
1. paired delta is negative	pass
2. interval excludes zero favourably	pass
3. no year degrades beyond tolerance	pass
4. survives nested calibration	pass
5. direction stable across seeds	pass
6. improvement exceeds integration error 10x	pass

6 of 6 checkable gate assertions hold. Passing them is not promotion. An interaction diagnostic across predeclared cohorts, reproduction from a clean identified commit, and confirmation on a future prospective period are not derivable from these artifacts. Production remains the five-seed Offset ensemble until they are met.

PER-SEED DELTA

Seed	Delta
7	-0.013234
17	-0.012556
42	-0.012215
123	-0.012210
256	-0.012571

5 of 5 seeds favour the candidate, and every seed agrees on the sign. Shown individually rather than only pooled, because a mean whose seeds disagree is a different claim from one whose seeds agree.

UTILITY SCALE

setting	value	note
2015	0.779697	fitted on strictly earlier years; fallback to the theoretical value
2016	1.389269	fitted on strictly earlier years
2017	1.453396	fitted on strictly earlier years
2018	1.439805	fitted on strictly earlier years
2019	1.450206	fitted on strictly earlier years
2020	1.428605	fitted on strictly earlier years
2021	1.426172	fitted on strictly earlier years
2022	1.420903	fitted on strictly earlier years
2023	1.428470	fitted on strictly earlier years
2024	1.426704	fitted on strictly earlier years
2025	1.423754	fitted on strictly earlier years
2026	1.426958	fitted on strictly earlier years

12 row(s) from the sensitivity artifact, in 1 study/studies. Where a study varies an integration setting, a result that moves with it is a property of the integration rather than of the model.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Standalone Probit against the market on the same races. It does not measure this model against the other three except through that shared yardstick.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.